

Day 7 content is complete. then provide day 8content

Day 8 — IPv4 Addressing & Subnetting

Today we'll go deeper into **IPv4 addressing and subnetting**. This is one of the most important networking interview topics because it connects directly to the **Layer 3 routing concepts** you learned in Days 2 and 7.

Day 8 Goal

By the end of today, you should be able to explain:

```
IPv4 Address
  ↓
Network Part + Host Part
  ↓
Subnet Mask
  ↓
CIDR Prefix
  ↓
Network Address
  ↓
Broadcast Address
  ↓
Usable Host Range
  ↓
Routing Decision
```

1. What is an IPv4 Address?

An IPv4 address is a **32-bit address**.

Example:

192.168.1.10

It contains four octets:

192	.	168	.	1	.	10
8 bits		8 bits		8 bits		8 bits

Total = 32 bits

Therefore:

$8 + 8 + 8 + 8 = 32 \text{ bits}$

2. Convert IPv4 to Binary

Example:

192.168.1.10

Each decimal number represents 8 bits.

192 = 11000000
168 = 10101000
1 = 00000001
10 = 00001010

So:

192.168.1.10

11000000.10101000.00000001.00001010

This is the actual 32-bit representation used by the networking system.

3. Why Do We Need a Subnet Mask?

Suppose:

IP = 192.168.1.10

Just knowing the IP isn't enough to determine which part represents:

Network

and which part represents:

Host

The **subnet mask** tells us where the network portion ends.

Example:

IP:
192.168.1.10

Mask:
255.255.255.0

Binary:

IP:
11000000.10101000.00000001.00001010

Mask:
11111111.11111111.11111111.00000000

The mask has:

24 ones + 8 zeros

Therefore:

Network = first 24 bits
Host = last 8 bits

4. CIDR Notation

Instead of writing:

192.168.1.10
255.255.255.0

we commonly write:

192.168.1.10/24

The /24 means:

The first 24 bits are the network prefix.

Visual:

192.168.1.10/24

<-----24 bits-----><--8 bits-->
NETWORK HOST

5. Common CIDR Prefixes

You should become comfortable with these:

CIDR	Subnet Mask	Host Bits
/8	255.0.0.0	24
/16	255.255.0.0	16
/24	255.255.255.0	8
/25	255.255.255.128	7
/26	255.255.255.192	6
/27	255.255.255.224	5
/28	255.255.255.240	4
/29	255.255.255.248	3
/30	255.255.255.252	2

For interviews, **/24 through /30** are particularly important.

6. Network Address

Let's take:

192.168.1.10/24

The network portion is:

192.168.1

The host portion is:

10

Therefore the network address is:

192.168.1.0

So:

Network:
192.168.1.0/24

7. Broadcast Address

For:

192.168.1.0/24

the host portion has 8 bits.

The broadcast address sets all host bits to 1 .

Therefore:

192.168.1.255

So:

Network Address = 192.168.1.0
Broadcast = 192.168.1.255

8. Usable Host Addresses

For:

192.168.1.0/24

the addresses are:

192.168.1.0
↓
Network address

192.168.1.1
↓
First usable host

...

192.168.1.254



Last usable host

192.168.1.255



Broadcast

Therefore the traditional usable host range is:

192.168.1.1



192.168.1.254

That's:

254 usable addresses

9. Why 254 and Not 256?

A /24 has:

$32 - 24 = 8$ host bits

Therefore:

$2^8 = 256$ total addresses

But traditionally:

1 address → Network

1 address → Broadcast

So:

$256 - 2 = 254$ usable host addresses

Formula

For ordinary IPv4 subnets:

Usable hosts = $2^{(\text{host bits})} - 2$

10. Example: /25

Now let's make it slightly harder.

192.168.1.0/25

/25 means:

25 network bits
7 host bits

Therefore:

$2^7 = 128$ addresses

Traditional usable hosts:

$128 - 2 = 126$

Subnet mask:

255.255.255.128

11. /25 Creates Two Subnets

A /24 network:

192.168.1.0/24

can be divided into two /25 networks.

192.168.1.0/25

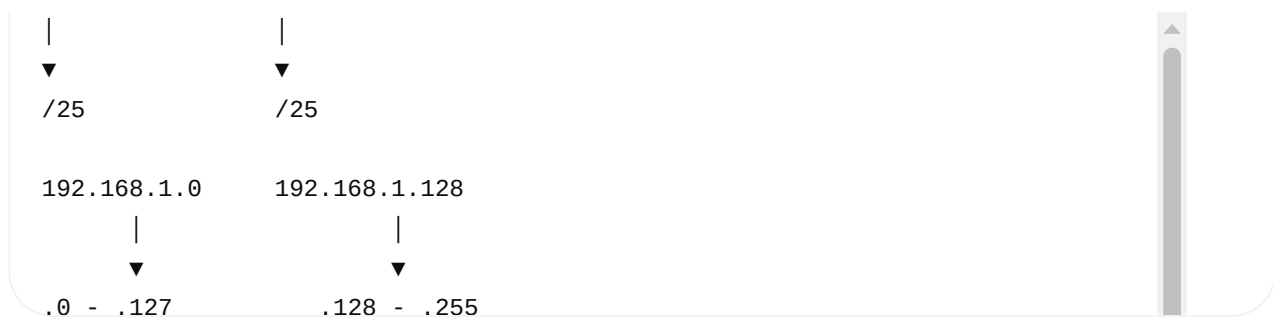
and

192.168.1.128/25

Visual:

192.168.1.0/24





12. First Subnet

192.168.1.0/25

Range:

192.168.1.0
↓
192.168.1.127

Therefore:

Network	= 192.168.1.0
First Host	= 192.168.1.1
Last Host	= 192.168.1.126
Broadcast	= 192.168.1.127

Usable:

126 hosts

13. Second Subnet

192.168.1.128/25

Range:

192.168.1.128
↓
192.168.1.255

Therefore:

Network = 192.168.1.128
First Host = 192.168.1.129
Last Host = 192.168.1.254
Broadcast = 192.168.1.255

14. /26

Now divide the /24 into /26 .

192.168.1.0/24
↓
/26

/26 gives:

$32 - 26 = 6$ host bits

Therefore:

$2^6 = 64$ addresses/subnet

Traditional usable:

$64 - 2 = 62$ hosts

15. /26 Creates Four Subnets

192.168.1.0/26
192.168.1.64/26
192.168.1.128/26
192.168.1.192/26

Visual:

192.168.1.0/24

├─ 0 - 63
├─ 64 - 127
├─ 128 - 191
└─ 192 - 255

Each subnet:

64 addresses
62 usable hosts

16. The "Block Size" Trick

This is extremely useful in interviews.

For:

255.255.255.192

look at the last octet:

$256 - 192 = 64$

Therefore the subnet boundaries occur every:

64

So:

0
64
128
192

This gives:

0 → 63
64 → 127
128 → 191
192 → 255

17. Example Interview Question

Question:

What subnet does:

192.168.1.70/26

belong to?

First:

/26

Mask:

255.255.255.192

Block size:

256 - 192 = 64

Subnet boundaries:

0
64
128
192

70 falls inside:

64 - 127

Therefore:

Network = 192.168.1.64

Broadcast:

192.168.1.127

Usable:

192.168.1.65
↓
192.168.1.126

18. Another Example

Find the network and broadcast for:

192.168.10.150/27

/27 :

Mask = 255.255.255.224

Block size:

256 - 224 = 32

Subnet boundaries:

0
32
64
96
128
160
192
224

150 falls in:

128 - 159

Therefore:

Network:
192.168.10.128

Broadcast:

192.168.10.159

Usable:

192.168.10.129
↓
192.168.10.158

19. Connect This to Day 7

This is where subnetting becomes practically important.

Suppose:

PC-A
IP = 192.168.1.10/24

and:

PC-B
IP = 192.168.1.20/24

PC-A calculates:

Network A = 192.168.1.0
Network B = 192.168.1.0

Same network.

Therefore:

```
PC-A
  |
  | ARP for PC-B
  ▼
MAC-B
  |
  ▼
Switch
  |
  ▼
PC-B
```

No router is required for the local delivery.

20. Different Network

Now:

PC-A
192.168.1.10/24

PC-B:

192.168.2.20/24

PC-A calculates:

PC-A network = 192.168.1.0
PC-B network = 192.168.2.0

Different networks.

Therefore PC-A sends the frame toward its **default gateway**.

```
PC-A
192.168.1.10
  |
  | Destination IP:
  | 192.168.2.20
  ▼
Default Gateway
192.168.1.1
  |
  ▼
Router
  |
  ▼
192.168.2.0 network
  |
  ▼
PC-B
192.168.2.20
```

This is why subnetting matters to routing.

21. The Router's Decision

Suppose router has:

```
Interface 1:
192.168.1.1/24
```

```
Interface 2:
192.168.2.1/24
```

PC-A:

```
192.168.1.10
```

Destination:

```
192.168.2.20
```

Router receives the packet.

It checks:

```
Destination IP
↓
```



Routing table
↓
192.168.2.0/24
↓
Interface 2

Then it forwards the packet toward PC-B.

22. Important: IP and MAC Again

This is one of the concepts you should now be able to explain clearly.

For different networks:

IP addresses

Generally remain the same end-to-end:

Source IP = PC-A
Destination IP = PC-B

MAC addresses

Change at every Layer-2 hop.

Example:

PC-A → Router

Src MAC = PC-A
Dst MAC = Router

Then:

Router → PC-B

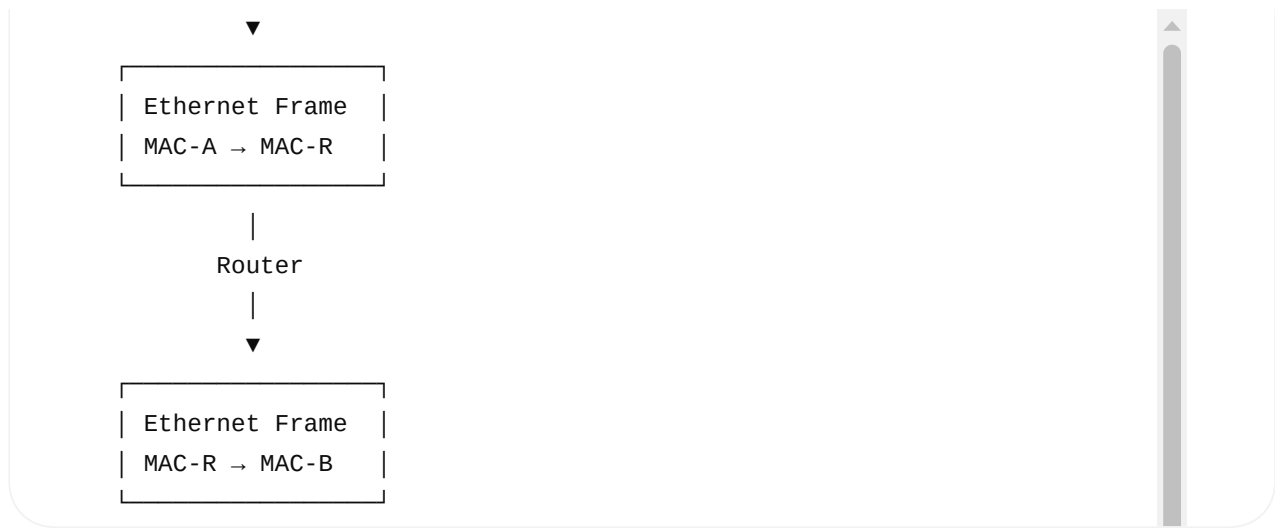
Src MAC = Router
Dst MAC = PC-B

Visual:

Layer 3

Src IP = PC-A
Dst IP = PC-B

stays
|



23. Private IPv4 Addresses

You should know these for interviews.

10.0.0.0/8

10.0.0.0 – 10.255.255.255

172.16.0.0/12

172.16.0.0 – 172.31.255.255

192.168.0.0/16

192.168.0.0 – 192.168.255.255

These are commonly used for private networks.

Example home network:

Router

192.168.1.1

Laptop

192.168.1.10

Phone

192.168.1.11

TV

192.168.1.20

24. Public vs Private IP

Private IP



Used inside private networks



Example:

192.168.1.10

Public IP



Globally routable address



Used for Internet communication

A home network commonly looks conceptually like:

Laptop

192.168.1.10



Home Router

Private IP: 192.168.1.1

Public IP: ISP-assigned



Internet

NAT, which we'll cover later, allows many private hosts to share public connectivity.

25. Special IPv4 Addresses You Should Know

Address/Range	Meaning
0.0.0.0	Unspecified / context-dependent
127.0.0.1	IPv4 loopback
127.0.0.0/8	Loopback range
169.254.0.0/16	IPv4 link-local
255.255.255.255	Limited broadcast

The most important:

```
127.0.0.1
```

↓

```
localhost / loopback
```

Example:

```
Application
```

|

```
| connect(127.0.0.1, 8080)
```

▼

```
Same machine
```

The traffic doesn't need to travel through the physical network like normal remote traffic.

26. Linux Commands for Day 8

Since your interview target involves Linux, know these.

View IP addresses

```
ip addr
```

or:

```
ip a
```

View routing table

```
ip route
```

View interface information

```
ip link
```

Example:

```
$ ip route
```

```
default via 192.168.1.1 dev eth0  
192.168.1.0/24 dev eth0
```

This tells you:

```
192.168.1.0/24
```

```
↓
```

```
directly connected network
```

```
default
```

```
↓
```

```
send other destinations to gateway
```

27. Very Important Routing Concept

Suppose:

```
Destination:  
192.168.1.50
```

Routing table:

```
192.168.1.0/24 dev eth0  
default via 192.168.1.1
```

Which route is selected?

```
192.168.1.0/24
```

Why?

Because it is a **more specific route** than the default route.

This leads to an important routing rule:

Longest-prefix match

28. Longest Prefix Match

Suppose routing table contains:

10.0.0.0/8
10.10.0.0/16
10.10.20.0/24
default

Destination:

10.10.20.50

Matches:

10.0.0.0/8
10.10.0.0/16
10.10.20.0/24

Which one wins?

10.10.20.0/24

because /24 is the longest/more-specific matching prefix.

Visual:

/8
└── matches

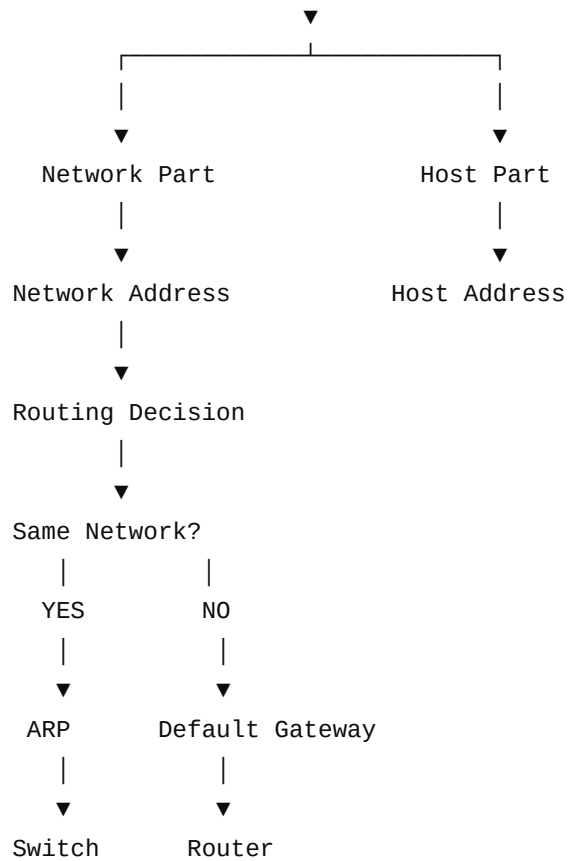
/16
└── matches

/24
└── matches
 ↑
 BEST

This is fundamental to router forwarding.

29. Day 8 Complete Mental Model

IPv4 Address
|
▼
32-bit address
|
▼
Subnet Mask
|



30. Day 8 Interview Questions

Before moving to Day 9, you should be able to answer these:

Basic

1. What is an IPv4 address?
2. How many bits are in IPv4?
3. What is an octet?
4. What is a subnet mask?
5. What does /24 mean?
6. What is CIDR?

Subnetting

7. What is the network address?
8. What is the broadcast address?
9. How many usable hosts are in /24 ?
10. How many usable hosts are in /26 ?
11. How many /26 subnets can be created from /24 ?
12. Find the network address of 192.168.1.70/26 .

13. Find the broadcast address of $192.168.1.70/26$.

14. Find the network address of $192.168.10.150/27$.

Routing

15. How does a host determine whether the destination is on the same network?

16. What happens when the destination is on a different network?

17. What is a default gateway?

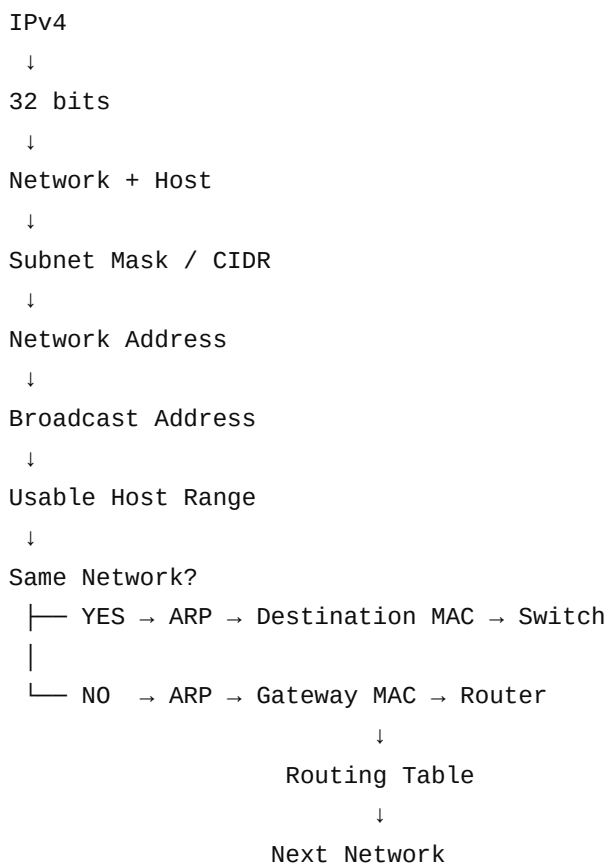
18. What is longest-prefix match?

19. Why does a router need a subnet mask/prefix?

20. What changes when a packet crosses a router: IP address or MAC address?

★ Day 8 Interview Summary

Memorize this chain:



The 5 formulas/concepts to remember

1. IPv4 = 32 bits

2. Host bits = 32 - prefix length

3. Total addresses = $2^{(\text{host bits})}$

4. Traditional usable hosts = $2^{(\text{host bits})} - 2$

5. Router forwarding = Longest Prefix Match

Day 8 is complete when you can calculate network address, broadcast address, host range, and determine whether PC-A sends directly to PC-B or to the default gateway.

Next in the fixed schedule: Day 9 — Routing Fundamentals + Routing Tables.

Today 7:54 AM

18. Another Example Find the network and broadcast for: /27: Block size: Subnet boundaries:
150 falls in: Therefore: Broadcast: Usable:

[explain clearly](#)

Absolutely. Let's slow down and understand **why** 192.168.10.150/27 **gives network** 192.168.10.128 **and broadcast** 192.168.10.159. This is a very important subnetting calculation for interviews.

Example: 192.168.10.150/27

We need to find:

1. Network address
2. Broadcast address
3. First usable IP
4. Last usable IP

Step 1 — Understand /27

IPv4 has **32 bits**.

32 bits total

/27 means:

27 bits → Network
5 bits → Host

Because:

$$32 - 27 = 5$$

So:

192.168.10.150/27

Network = 27 bits	Host
	5 bit

Step 2 — Find the subnet mask

/27 means 27 1 s followed by 5 0 s.

11111111.11111111.11111111.11100000

Convert each octet:

11111111 = 255

11111111 = 255

11111111 = 255

11100000 = 224

Therefore:

/27 = 255.255.255.224

Step 3 — Why do we calculate 256 - 224 ?

The interesting part is the **last octet**.

Our mask is:

255.255.255.224

↑
important octet

Calculate:

$$256 - 224 = 32$$

This 32 is called the **block size**.

It tells us:

Each /27 subnet contains 32 IP addresses.

Step 4 — Create the subnet boundaries

If every subnet has **32 addresses**, start from 0 and keep adding 32:

```
0
0 + 32 = 32
32 + 32 = 64
64 + 32 = 96
96 + 32 = 128
128 + 32 = 160
160 + 32 = 192
192 + 32 = 224
```

So the subnet boundaries are:

```
0
32
64
96
128
160
192
224
```

These are the **network addresses**.

Therefore:

```
192.168.10.0/27
192.168.10.32/27
192.168.10.64/27
192.168.10.96/27
192.168.10.128/27
192.168.10.160/27
192.168.10.192/27
192.168.10.224/27
```

Step 5 — Where does 150 belong?

Our IP is:

192.168.10.150

We only need to look at the last octet:

150

Look at our ranges:

0	——	31	
32	——	63	
64	——	95	
96	——	127	
128	——	159	← 150 is here
160	——	191	
192	——	223	
224	——	255	

150 is between:

128 and 159

Therefore, the subnet containing 150 starts at:

128

So:

Network Address = 192.168.10.128

Step 6 — Find the broadcast address

Our subnet is:

128 — 159

The **last address** in that subnet is the broadcast address.

Therefore:

Broadcast = 192.168.10.159

That's it.

Step 7 — Find usable host addresses

Remember:

Network address



cannot normally be assigned to a host

Broadcast address



cannot normally be assigned to a host

Our subnet:

192.168.10.128



Network

192.168.10.129



First host

...

192.168.10.158



Last host

192.168.10.159



Broadcast

Therefore:

Network:

192.168.10.128

First usable:

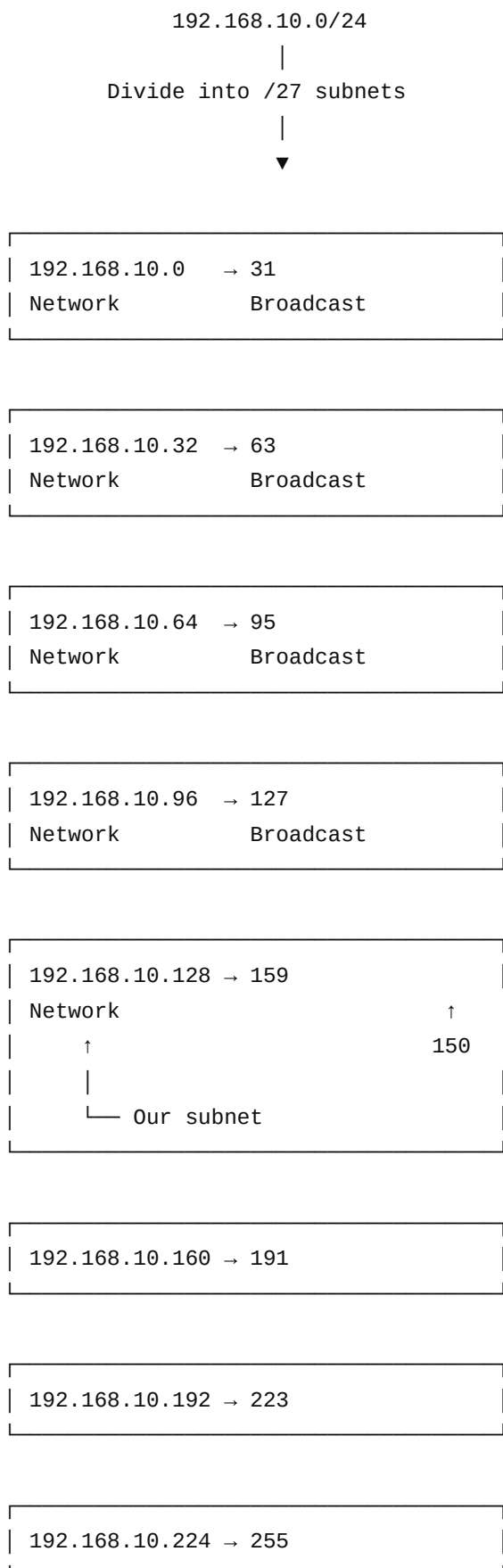
192.168.10.129

Last usable:

192.168.10.158



Visualize the Complete Subnet



The Most Important Trick

For subnetting questions like:

192.168.10.150/27

you don't need to convert everything to binary every time.

Use this **4-step interview method**:

① Find mask

/27

↓

255.255.255.224

② Calculate block size

256 - 224 = 32

③ Find the range containing 150

0-31

32-63

64-95

96-127

128-159 ← 150

160-191

192-223

224-255

④ Answer

Network = 192.168.10.128

Broadcast = 192.168.10.159

Usable:

192.168.10.129 - 192.168.10.158

Why is 160 NOT the network for 150 ?

This is a common confusion.

You might think:

150 is closer to 160

But that's not how subnet boundaries work.

The boundaries are fixed:

0 → 32 → 64 → 96 → 128 → 160 → 192 → 224

150 comes **after 128 but before 160**:

128	—————	150	—————	159
↑		↑		↑
Network		Host		Broadcast

Therefore:

Network = 128

not 160 .

One More Example

Try:

192.168.10.200/27

We already know /27 gives blocks of 32 :

0-31
32-63
64-95
96-127
128-159
160-191
192-223 ← 200 is here
224-255

Therefore:

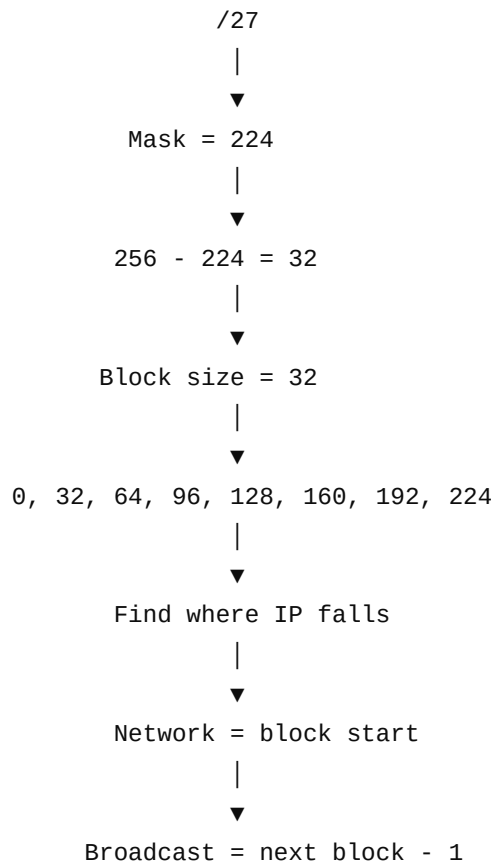
Network = 192.168.10.192
Broadcast = 192.168.10.223



First = 192.168.10.193
Last = 192.168.10.222



★ Remember This Pattern



So for:

192.168.10.150/27

the answer is:

Network	:	192.168.10.128
First	:	192.168.10.129
Last	:	192.168.10.158
Broadcast	:	192.168.10.159

This block-size method is the one I recommend you use first in interviews. Once this is comfortable, binary calculation becomes much easier to understand rather than something you have to memorize.